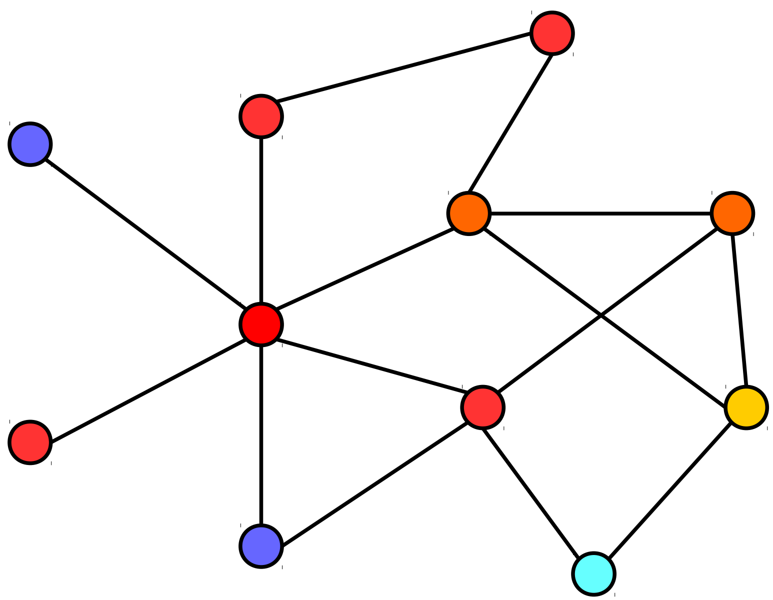
Toward an uncertainty principle for weighted graphs

Bastien Pasdeloup*, Réda Alami*, Vincent Gripon*, Michael Rabbat**

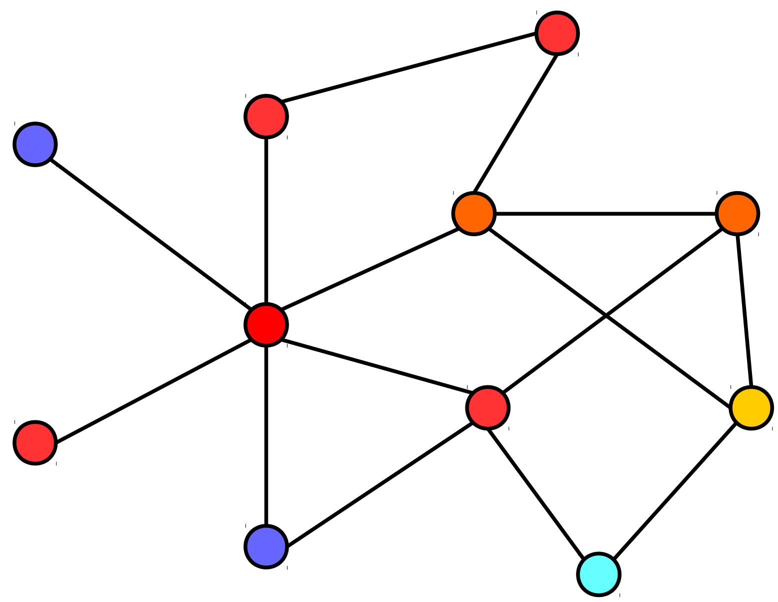
* *name.surname@telecom-bretagne.eu*
** *name.surname@mcgill.ca*



$$\mathbf{x} = \begin{pmatrix} -0.5 \\ 0.5 \\ 0.4 \\ 0.45 \\ 0.7 \\ 0.3 \\ -0.45 \\ 0.4 \\ 0.25 \\ -0.3 \\ 0.1 \end{pmatrix}$$

Generalization of classical Fourier analysis to more complex domains

Normalized Laplacian: $\mathbf{Ł} \triangleq \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\mathbf{W}\mathbf{D}^{-\frac{1}{2}}$



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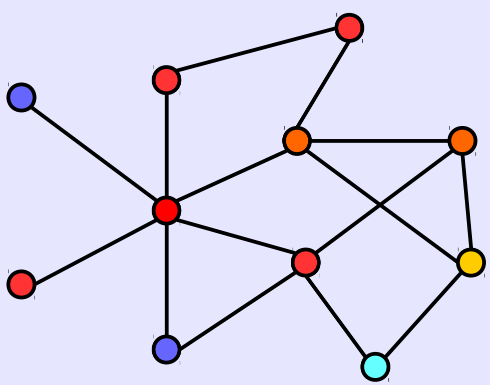
<u>Assumption:</u> normalized signals

Generalization of classical Fourier analysis to more complex domains

<u>Normalized Laplacian:</u> $Ł \triangleq \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\mathbf{W}\mathbf{D}^{-\frac{1}{2}}$

Graph Fourier transform

$$\mathcal{G} = \langle N, \mathbf{W} \rangle$$



Graph domain

Diagonalization of Ł

(Ł real and symmetric)

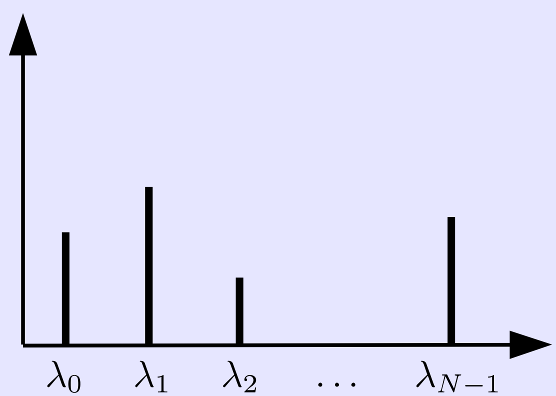
Graph Fourier transform

$$\hat{\mathbf{x}} = \mathcal{X}^\top \mathbf{x}$$

Inverse graph Fourier transform

$$\mathbf{x} = \mathcal{X}\hat{\mathbf{x}}$$

$$\langle \mathcal{X}, \mathbf{\Lambda} \rangle$$



Spectral domain

A portage of tools

$$(\mathbf{f} * \mathbf{h})(i) := \sum_{l=0}^{N-1} \hat{\mathbf{f}}(\lambda_l)\hat{\mathbf{h}}(\lambda_l)\mathbf{u}_l(i)$$

Convolution

$$(\hat{D}_s\mathbf{g})(\lambda) := \hat{\mathbf{g}}(s\lambda)$$

Dilatation

$$\hat{\mathbf{f}}_{\mathbf{out}}(\lambda_l) = \hat{\mathbf{f}}_{\mathbf{in}}(\lambda_l)\hat{\mathbf{h}}(\lambda_l)$$

Filtering

$$(T_n\mathbf{g})(i) := \sqrt{N}(\mathbf{g} * \delta_n)(i)$$

Translation

$$(M_k\mathbf{g})(i) := \sqrt{N}\mathbf{u}_k(i)\mathbf{g}(i)$$

Modulation

...

Reference paper: Shuman *et. al* – *The emerging field on signal processing on graphs* – 2013

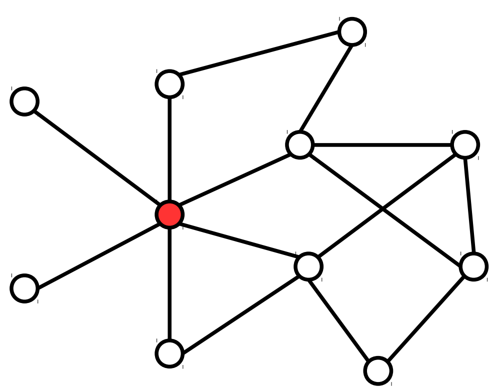
The tool that interests us here

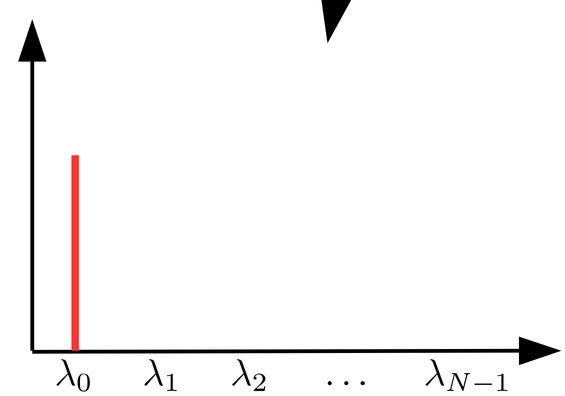
Heisenberg principle

A signal cannot be both fully localized in the **time** and **frequency** domains

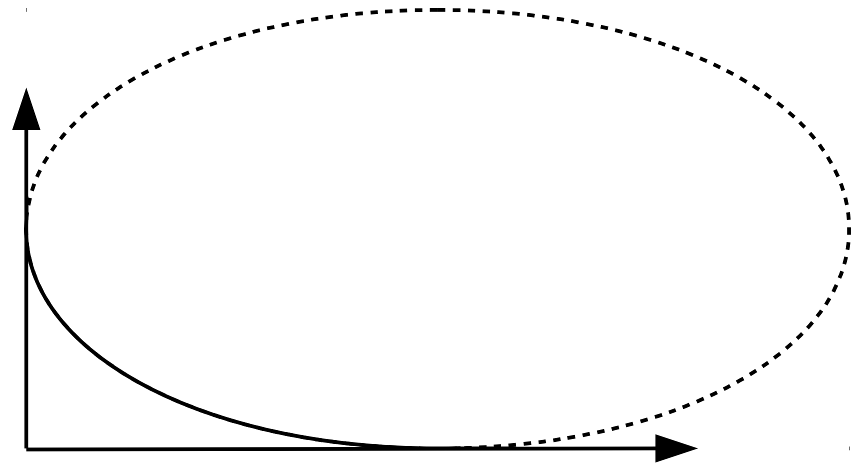
On graphs

A signal cannot be both fully localized in the **graph** and **spectral** domains





$$\Delta^2_{\mathcal{G},u_c}(\mathbf{x}) = \sum_{n=0}^{N-1} d(u_c, u_n)\mathbf{x}_n^2$$

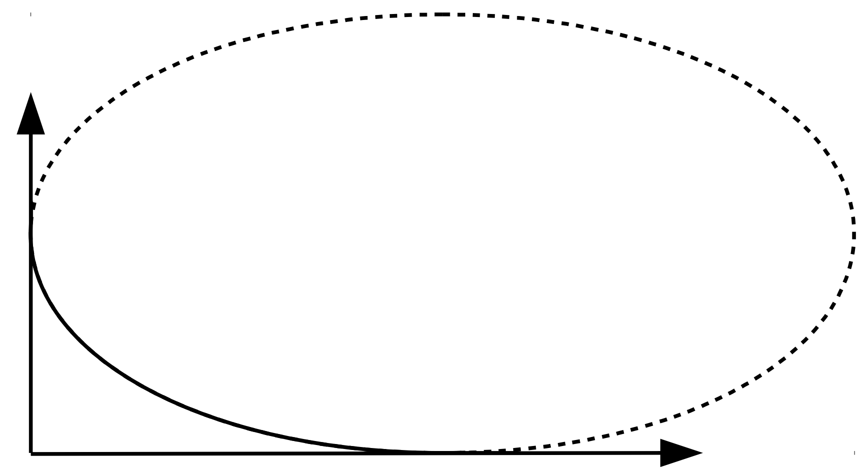


$$\Delta^2_s(\mathbf{x}) = \sum_{n=0}^{N-1} \lambda_n \hat{\mathbf{x}}_n^2$$

Reference paper: Agaskar & Lu – *A spectral graph uncertainty principle* – 2013

The uncertainty principle applied to graphs

$$\boxed{\Delta^2_{\mathcal{G},u_c}(\mathbf{x}) = \sum_{n=0}^{N-1} d_{geo}(u_c, u_n)\mathbf{x}_n^2}$$



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Graph spread: – Around a particular node
– Use of the geodesic (*i.e* shortest path) distance

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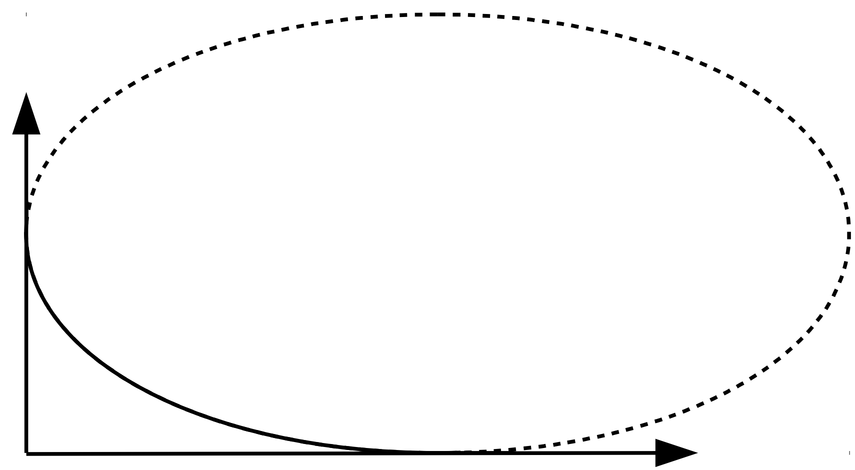

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Spectral spread: – Around the smallest eigenvalue (=*0*)
– This choice is often discussed, but corresponds to the steady state

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0 diffusion steps

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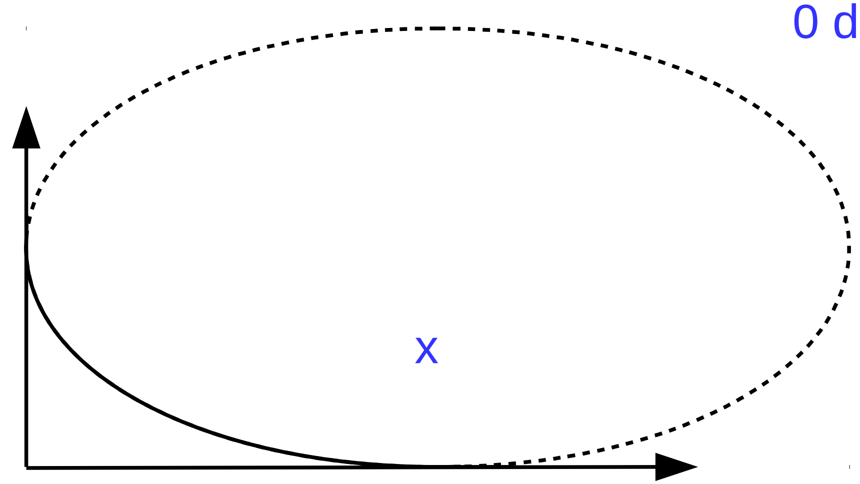

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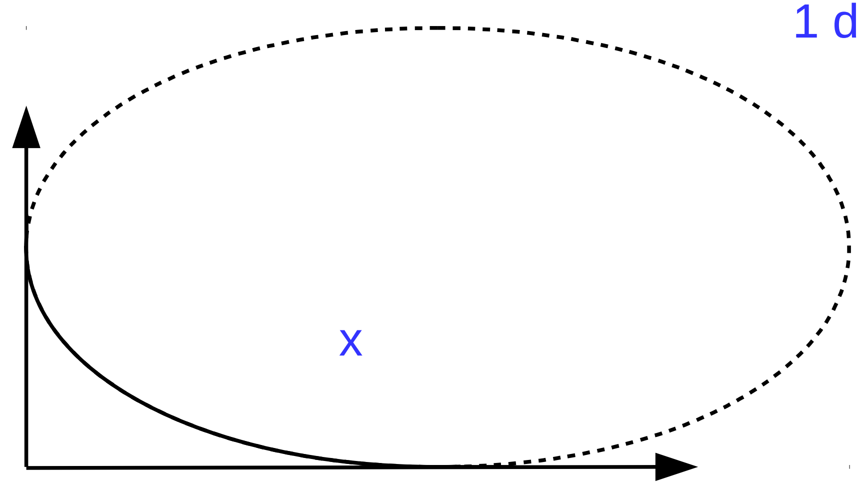

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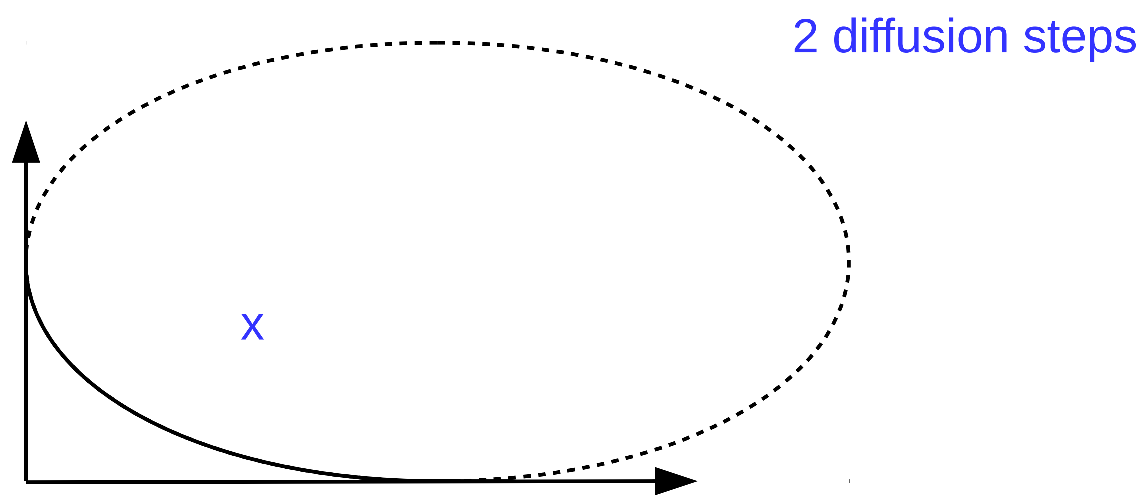

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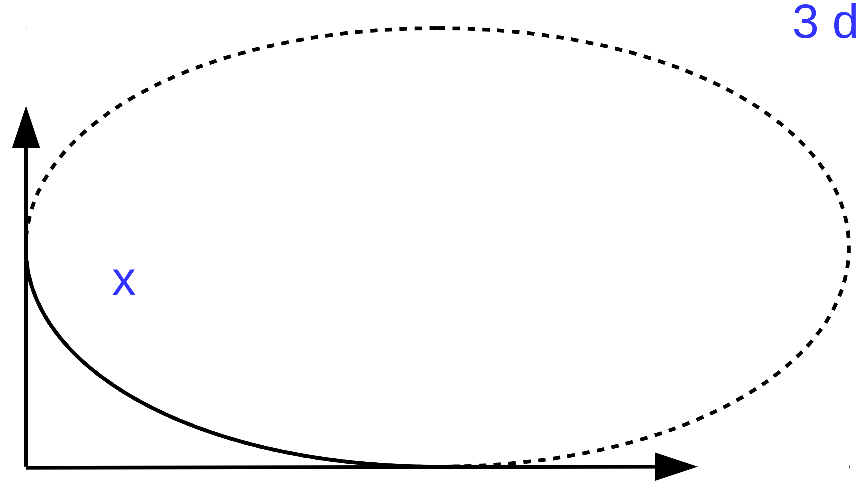

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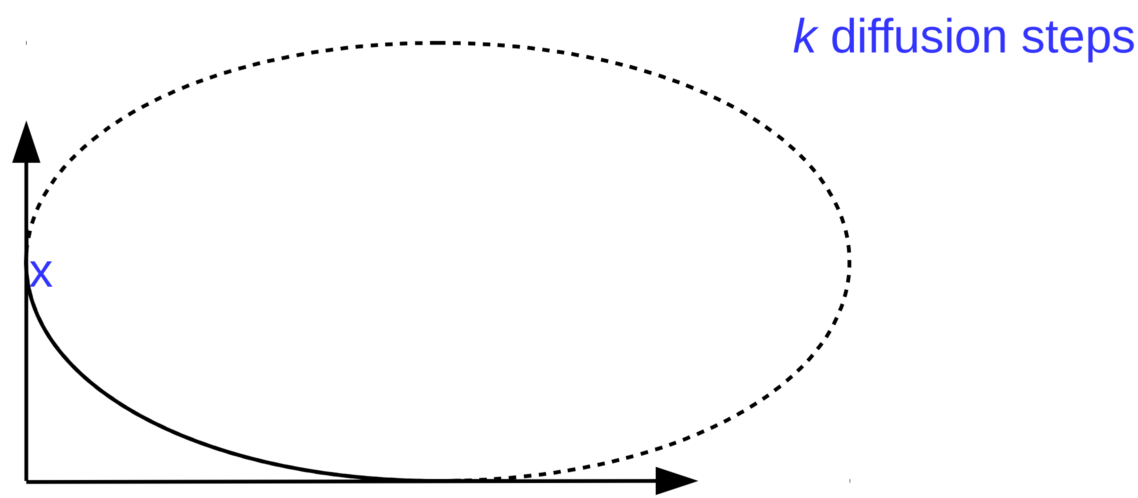

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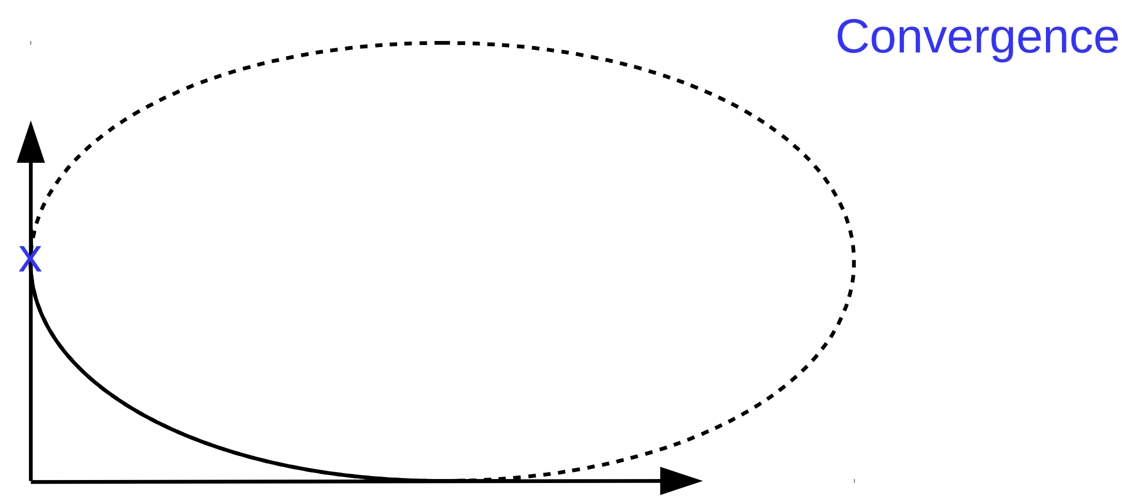

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Convergence

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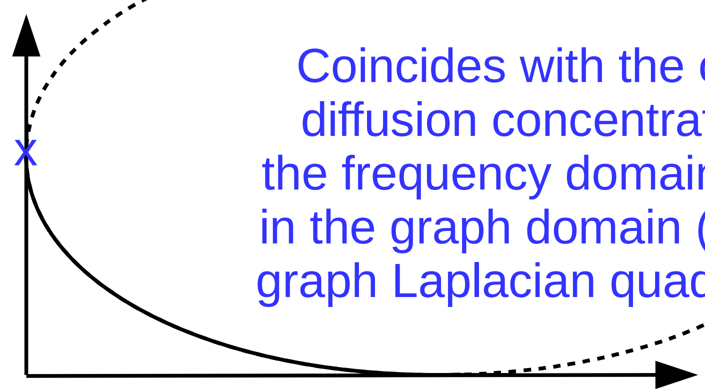

Coincides with the observation that diffusion concentrates the signal in the frequency domain, but smoothes it in the graph domain (*i.e* decreases the graph Laplacian quadratic form: $\mathbf{x}^\top \mathcal{L}\mathbf{x}$)

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Current limitations of their work

$$\Delta^2_{\mathcal{G},u_c}(\mathbf{x}) = \sum_{n=0}^{N-1} d_{geo}(u_c, u_n)\mathbf{x}_n^2$$

– Currently addresses unweighted graphs only

– Arbitrarily uses the geodesic distance

– Does not give any hint on the choice of the reference node

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Addressed in this paper

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A naive extension to weighted graphs

Idea #1: Let's do the same with a weighted adjacency matrix!

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Geodesic distance is also defined for weighted graphs

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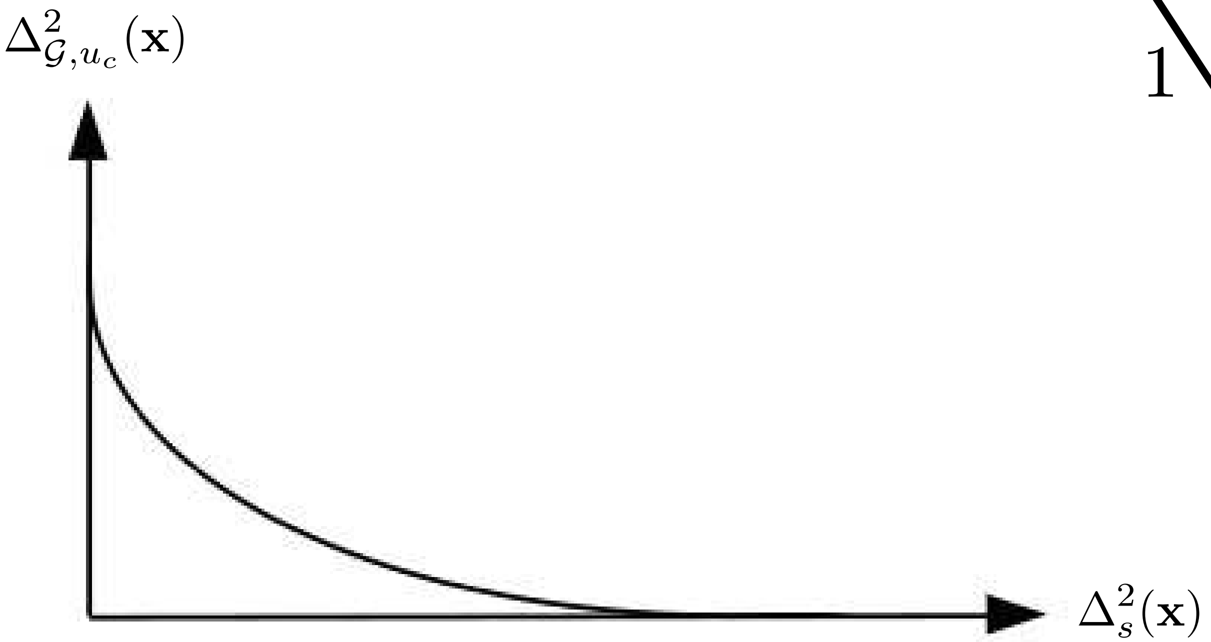
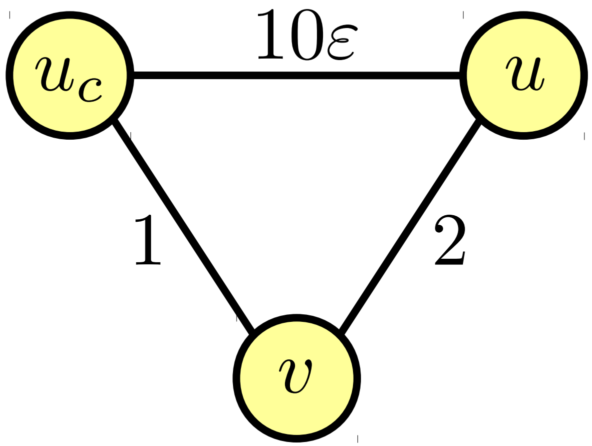
$\Rightarrow$ This leads to a discontinuity of the graph spread with respect to the graph structure

Illustration of the problem





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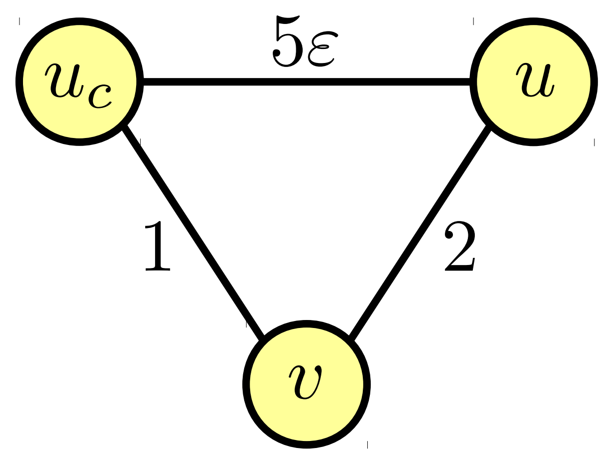

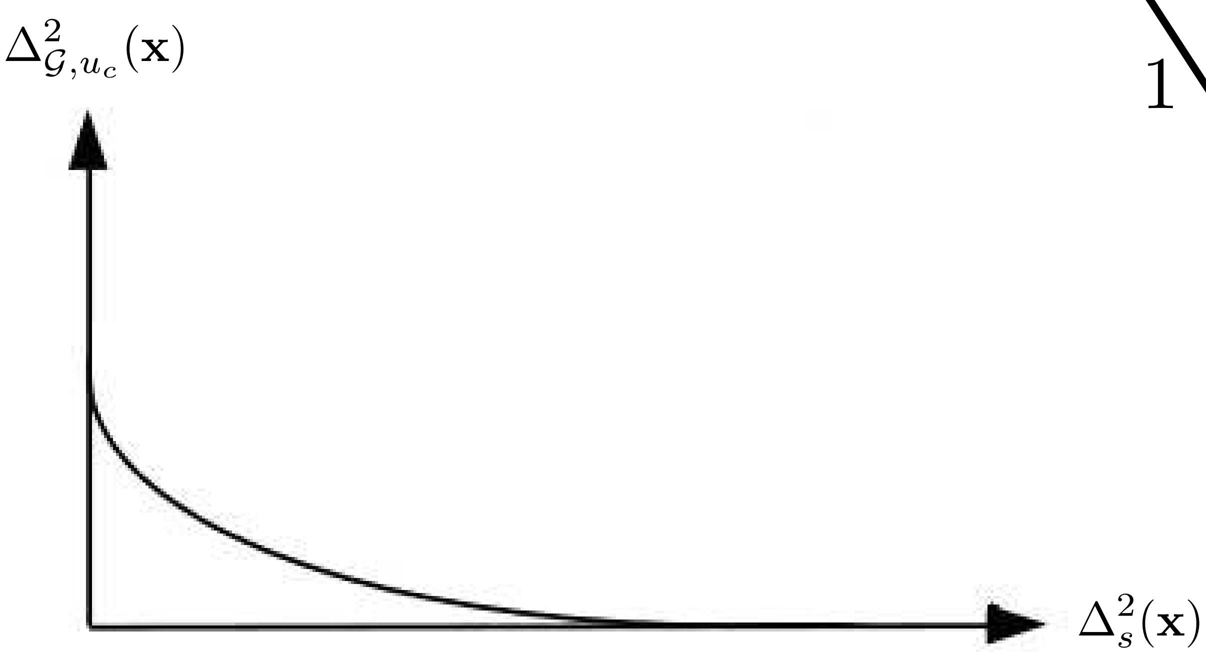

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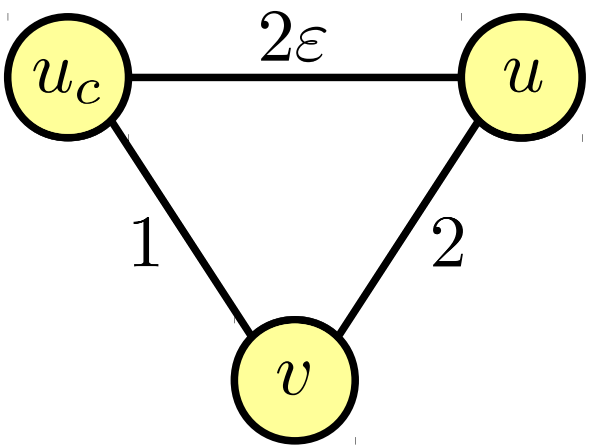

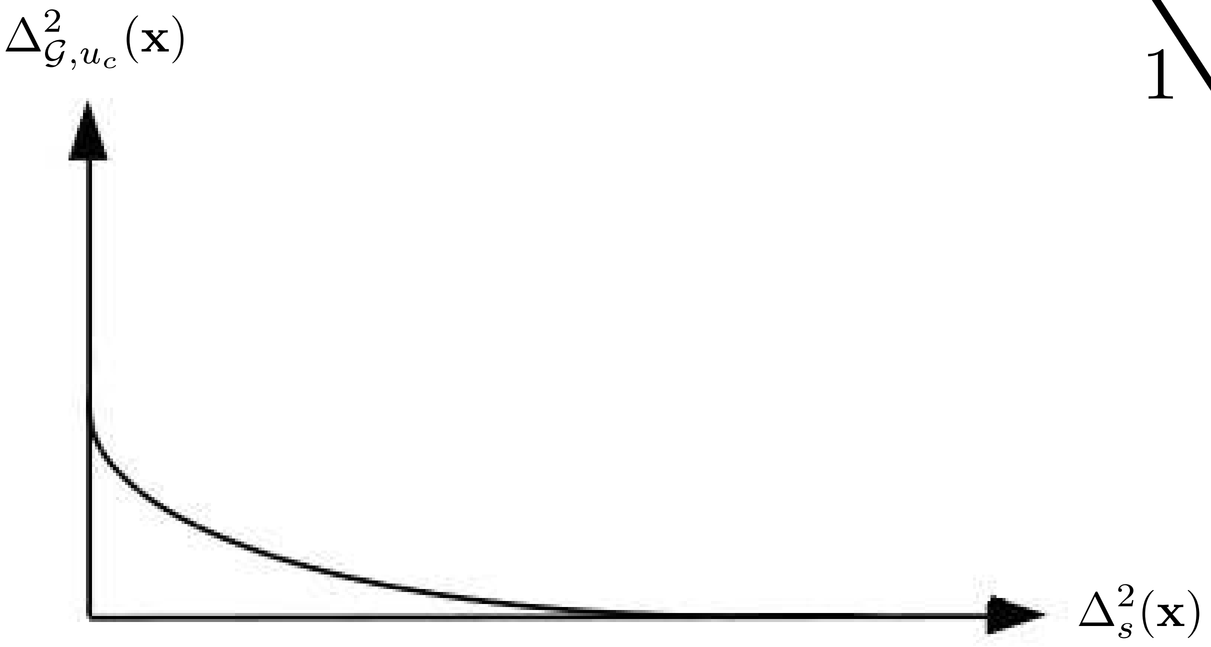

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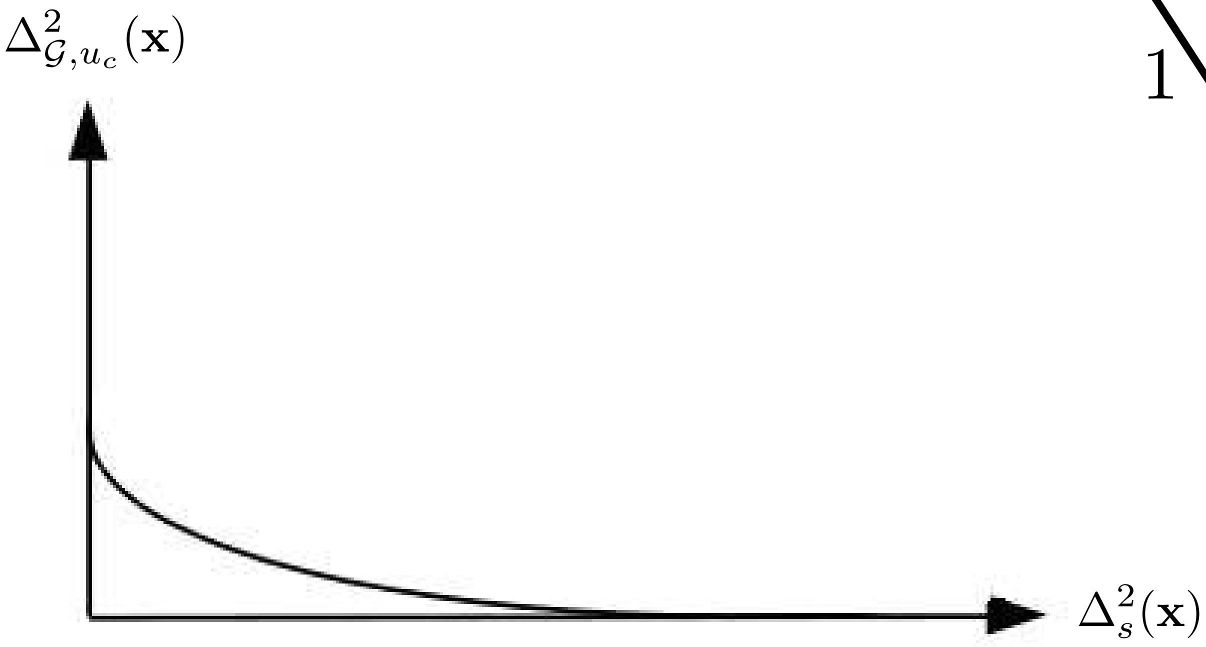

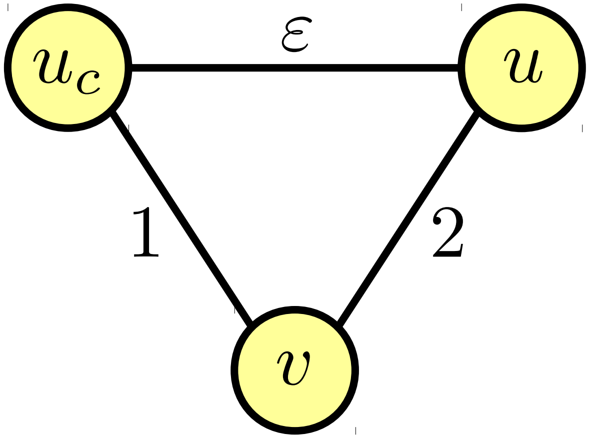

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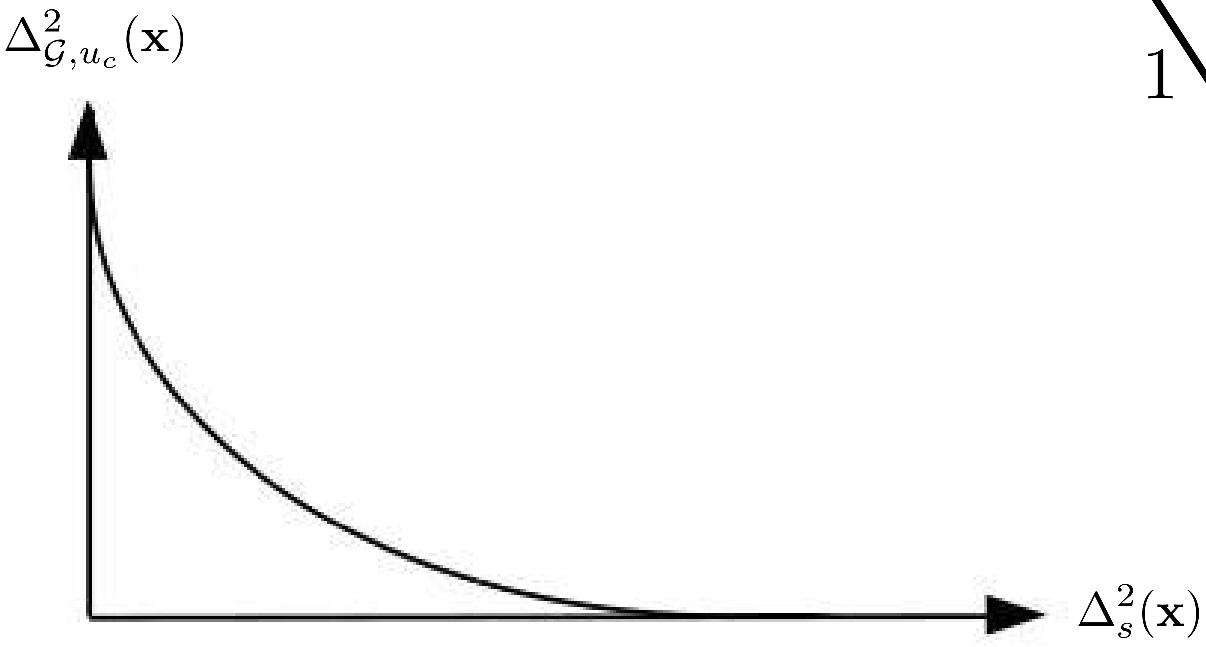
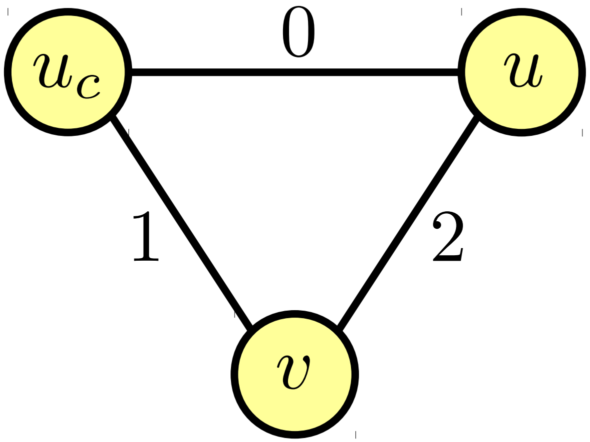
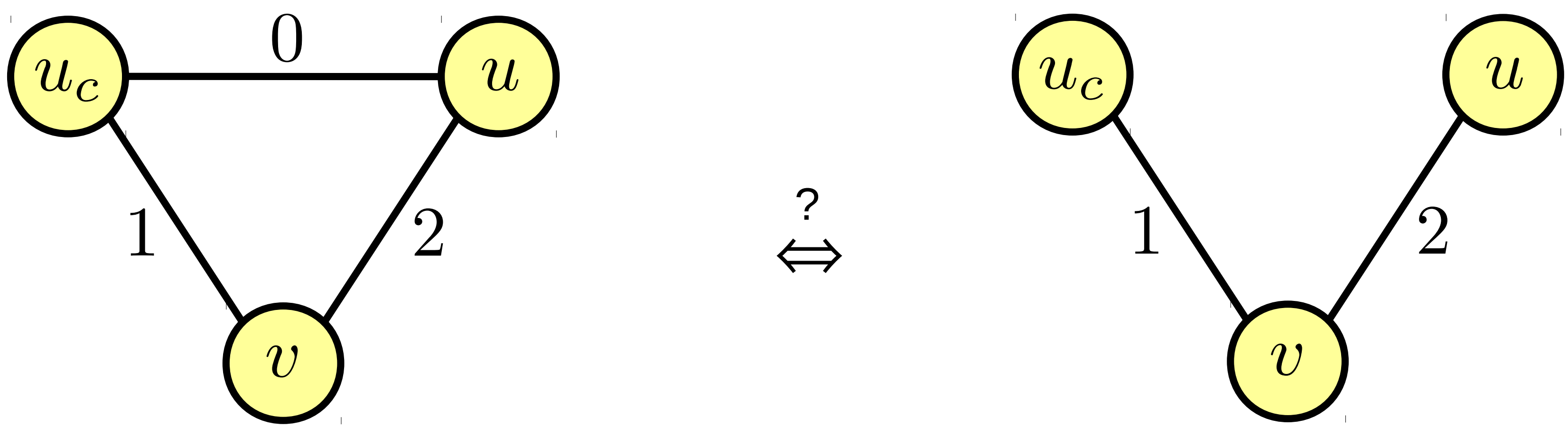




Explanation of the discontinuity



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From now on, $\mathbf{W}$ will be regarded as a similarity matrix

A change of distance function

New definition of graph spread: $\Delta^2_{\mathcal{G},u_c}(\mathbf{x}) = \sum_{n=0}^{N-1} d(u_c, u_n)\mathbf{x}_n^2$

– Similar to the previous definition

– We need to change the distance function so that we do not use $\mathbf{W}$ as distances

Proposed requirements on the distance function d:

1°) $\forall u, v \in \mathcal{V} : d(u, v) \geq 0$

2°) $\forall u, v \in \mathcal{V} : d(u, v) = 0 \Leftrightarrow u = v$

3°) d is continuous, and if we increase $\mathbf{W}_{u,v}$ for a single edge (u, v), then $\forall u', v' \in \mathcal{V} : d(u', v')$ does not increase

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Similar graphs should have similar spreads

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Proposed requirements on the distance function d:

1°) $\forall u, v \in \mathcal{V} : d(u,v) \geq 0$ The spread should be a positive quantity

2°) $\forall u, v \in \mathcal{V} : d(u,v) = 0 \Leftrightarrow u = v$ To have $\Delta^2_{\mathcal{G},u_c}(\mathbf{x}) = 0$ for and only for a signal localized on u_c

3°) d is continuous, and if we increase $\mathbf{W}_{u,v}$ for a single edge (u,v), then $\forall u', v' \in \mathcal{V} : d(u', v')$ does not increase

Similar graphs should have similar spreads

$\mathbf{W}$ is considered as a matrix of similarities

Inverse similarity matrix: $\forall u, v \in \mathcal{V} : \bar{\mathbf{S}}_{u,v} \triangleq \begin{cases} \infty & \textbf{if } \mathbf{W}_{u,v} = 0 \\ 0 & \textbf{if } u = v \\ \frac{1}{\mathbf{W}_{u,v}} & \textbf{otherwise} \end{cases}$

– Use of the squared geodesic distance with $\bar{\mathbf{S}}$ instead of $\mathbf{W}$: $d^2_{geo(\bar{\mathbf{S}})}$

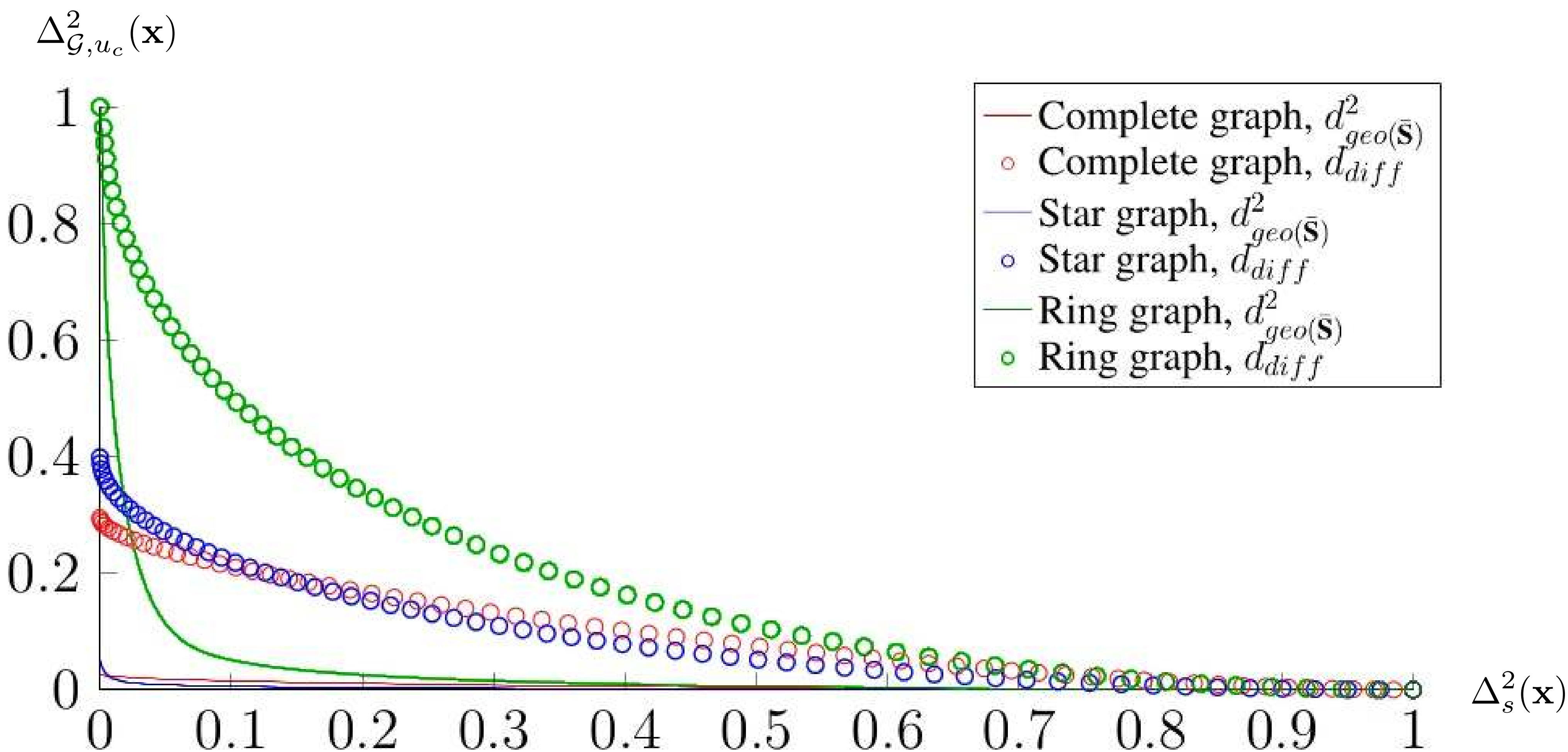
– Simple correction of the discontinuity

– Basically, $x \mapsto \dfrac{1}{x}$ can be replaced by any positive non-increasing function (*eg.* Gaussian kernel)

Diffusion distance: $\forall u, v \in \mathcal{V} : d_{diff} \triangleq \|(\mathbf{I} + \alpha \mathbf{L})^{-1}(\mathbf{x}_u - \mathbf{x}_v)\|$

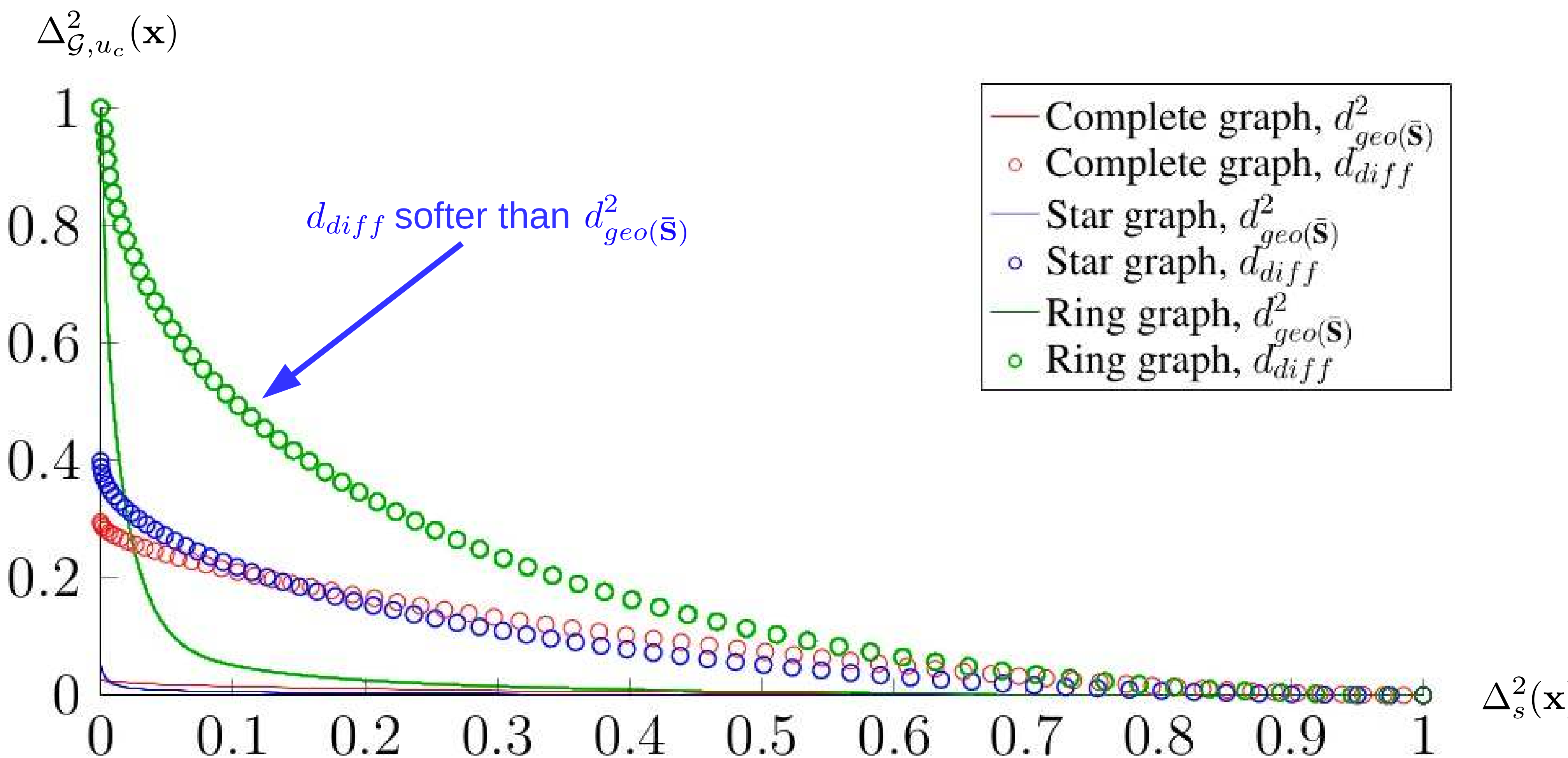
– $\mathbf{x}_u$ is a signal fully localized on node u

Associated uncertainty curves



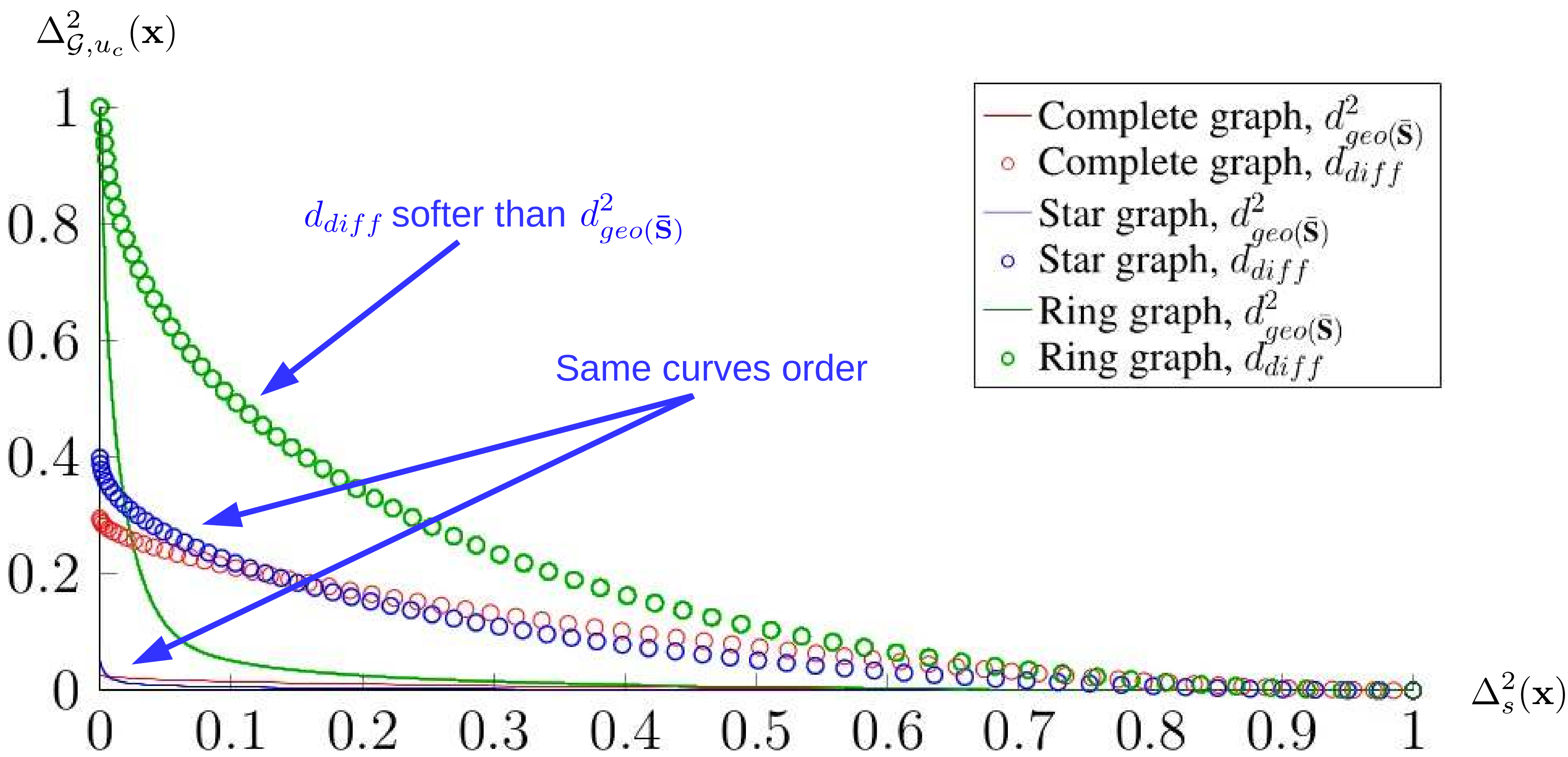
Mean uncertainty curves for various graphs, for 100 random weights

Associated uncertainty curves



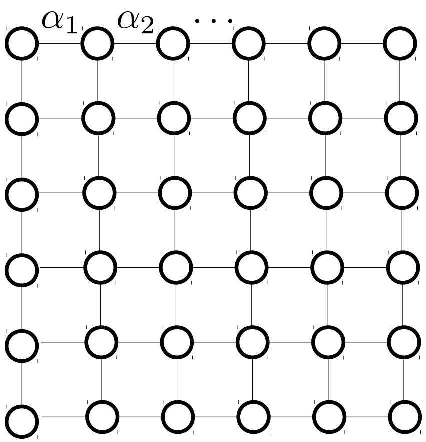
Mean uncertainty curves for various graphs, for 100 random weights

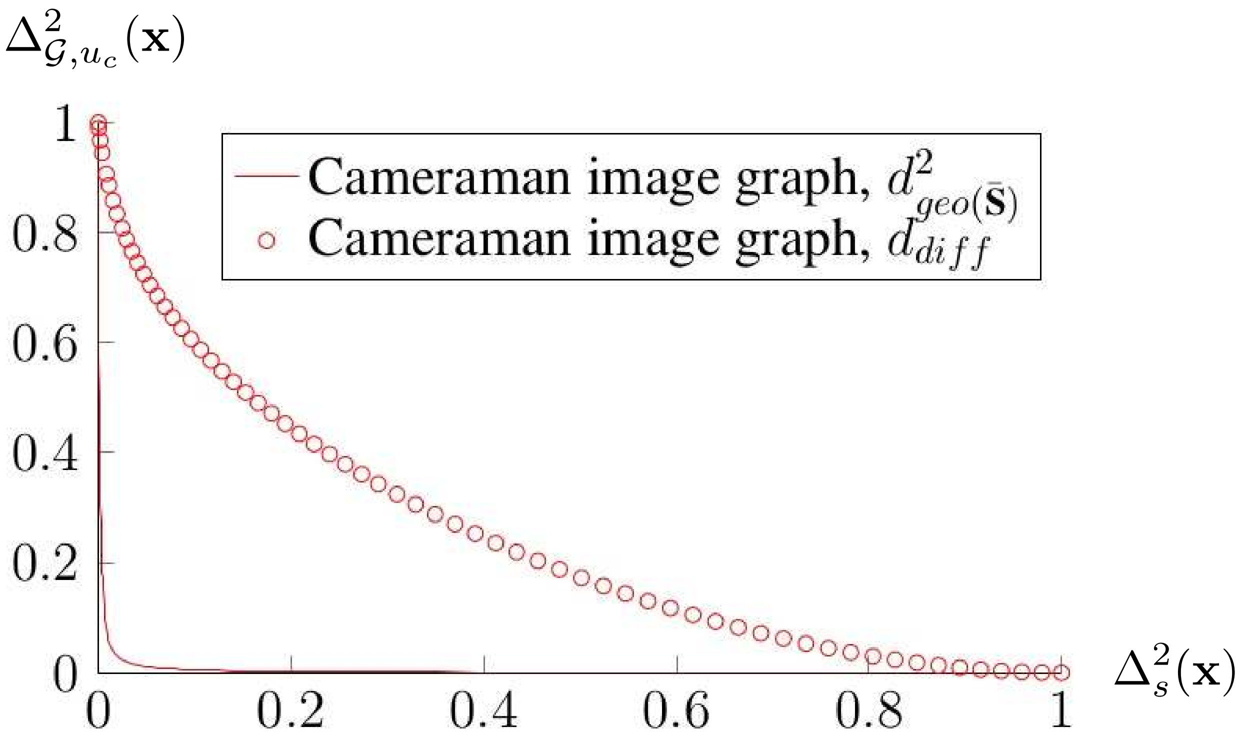
Associated uncertainty curves



Mean uncertainty curves for various graphs, for 100 random weights

Application to semi-localized graphs





Summary of the paper:

– Extension of the uncertainty principle to weighted graphs

– Highlight of the confusion when using $\mathbf{W}$ as a distances matrix

– Statement of requirements on the distance functions used in $\Delta^2_{\mathcal{G},u_c}(\mathbf{x})$

Future work:

– Study of the impact of the choice for the distance function

– Obtention of the analytical expressions of the curves

– Study of the impact of the choice of the central node (*eg.* to compare graphs)

– Use the uncertainty principle to guide a graph reconstruction process

Toward an uncertainty principle for weighted graphs

Bastien Pasdeloup*, Réda Alami*, Vincent Gripon*, Michael Rabbat**

* *name.surname@telecom-bretagne.eu*
** *name.surname@mcgill.ca*